

Technical Document #562: Software Engineering - Comprehensive Overview

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Design patterns provide reusable solutions to common software design problems. Singleton, factory, and observer patterns are widely used.

Continuous integration (CI) automatically builds and tests code changes. It catches integration issues early in the development process.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

Continuous deployment (CD) automatically deploys code changes to production. It enables rapid, reliable releases.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Test-driven development (TDD) writes tests before writing code.
- API design defines how software components communicate.
- Continuous integration (CI) automatically builds and tests code changes.